

Chapter 158 — Active Management Strategies for Localised Prostate Cancer

- **Prostate-specific antigen (PSA) era uniformity:** PSA testing and community screening from the early 1990s → radical treatment for most patients regardless of risk, apparently vindicated by serially falling US prostate cancer mortality.
- **Evidence against treating everyone:**
 - Observation of low-risk disease → **low rates of metastasis and prostate cancer death.**
 - Randomised treatment-versus-observation trials → mortality reduction **too small to justify the quality-of-life cost**, largely because of high early death from other causes.
 - Large randomised screening trials → **no robust mortality decline for all patients** despite prolonged follow-up.
- **15-year randomised data in favourable-risk disease:** surveillance → more metastatic progression but **no difference in prostate cancer mortality**; explained by high crossover to treatment within 5–10 years, competing mortality and much-improved treatment of metastatic disease.
- **The shift:** from choosing a treatment → also choosing its **timing relative to disease risk and expected longevity**; for favourable risk, early radical therapy may cost more quality of life than the cancer costs life.
- **Many early-stage patients still carry significant lifetime mortality risk** → contemporary care is individualised: longevity, disease aggressiveness and extent, patient goals.
- **Management spectrum:**
 - More likely to die of competing risks → **observation.**
 - Good longevity, benefit of treatment unclear → **active surveillance**, deferring therapy until need is clear.
 - Needing therapy → **radical prostatectomy (RP), external radiation, interstitial radiation, or thermal ablation (whole or partial gland).**
- **Modality selection** requires counselling on risks and benefits, recognition of host factors, comparative data and patient preference.

Concept — Management of localised prostate cancer is **a question of timing as well as modality**: cancer mortality risk is always weighed against competing mortality, and every later decision follows from that balance.

HOST EVALUATION

Longevity Assessment/Competing Risks for Mortality

- **Competing mortality dilutes treatment benefit** in large efficacy trials; several randomised trials show other-cause death risk exceeding prostate cancer death risk.
- **National Comprehensive Cancer Network (NCCN):** "life expectancy estimation is critical to informed decision making" in early detection and treatment.
- **Prostate Cancer Outcomes Study (PCOS)** (US prospective cohort): other-cause mortality modelled on comorbidity and age; **10-year prostate cancer mortality 3–18% by risk category versus ≥33% other-cause mortality with any comorbidity.**
- **The "10-year life expectancy" rule:** applied bluntly, it has produced lower treatment rates in older patients and possibly overtreatment of younger ones; best practice is an **individualised assessment of age, comorbidity and cancer mortality risk.**

- **Tools:** NCCN recommends **Social Security Administration tables or World Health Organization (WHO) Life Tables**; prostate cancer-specific models also exist, in which the age-comorbidity relationship is central.
- **Longevity is an indirect measure of comorbidity** and shapes both *whether* and *which* treatment: significant cardiac, pulmonary or haematological comorbidity may leave longevity and benefit intact but **contraindicate surgery (anaesthetic or bleeding risk)**.
- **Charlson Comorbidity Index (CCI):** most widely used scale; designed to predict **1-year inpatient mortality**; later shown prognostic across many cancers.
- **Prostate cancer-specific CCI (PCCI):** reweights comorbidities to predict **10-year mortality** in prostate cancer; in the Veterans Health Administration (VHA), age-adjusted 10-year other-cause mortality **rises steeply across score bands, approaching certainty at the highest scores**.
- **Care Assessment Needs (CAN) score:** automated electronic health record score (VHA) built for short-term hospitalisation and mortality; **stronger predictor of 5-year overall survival than CCI or PCCI** in both localised and metastatic disease.
- **Age-based bias:** clinicians judge longevity by age rather than comorbidity → comorbid younger men treated more often than healthy older men despite far higher other-cause mortality; treatment of patients with limited life expectancy falls with age; older high-risk patients undertreated despite reasonable 10-year longevity. Remedy: **age-adjusted longevity from validated comorbidity indices and life tables**.
- **Longevity must be weighed against aggressiveness:**
 - Aggressive cancers may benefit from treatment **despite shortened longevity**, especially at advanced stage.
 - VHA practice: RP use falling in low-risk and rising in high-risk disease, including in men with ≤10-year longevity.
 - High-risk, poorly differentiated disease: absolute benefit of aggressive treatment **shrinks and number needed to treat rises steeply** with increasing Charlson score.
 - Lead time to symptoms and death is governed by **Gleason score and stage**.
- **High risk of metastasis or death within 10 years → consider aggressive therapy** even with competing risks in the same period.

Figure 158.1 — Other-cause mortality by comorbidity index

- **Kaplan-Meier non-prostate-cancer survival** separates monotonically across original CCI bands (0, 1, 2, 3–4, 5+) and reweighted bands (0, 1–2, 3–5, 6–8, 9+); the **reweighted index better identifies the highest-risk group** (Daskivich et al., 2011).

Concept — Longevity matters only *relative to the cancer's own clock*: a competing risk argues against treatment only when it will act before the cancer does.

BASELINE URINARY/SEXUAL/BOWEL FUNCTION

Bladder Function

- **Baseline function sets the size of the quality-of-life benefit and risk**; pre-existing dysfunction → more intense or protracted morbidity. Assess with **validated instruments** for shared decision-making.
- **Instruments:** **International Prostate Symptom Score (IPSS)** (most common) and **University of California, Los Angeles (UCLA) Prostate Cancer Index**; analyse individual IPSS items to characterise symptoms.

Baseline state	Surgery	Radiation
Obstructive lower urinary tract symptoms (LUTS) from benign prostatic hyperplasia (BPH)	Improves symptoms , durably, without the usual age-related progression; severe baseline → more early postoperative bother	↑ Retention risk , clearly after brachytherapy; not clearly after external-beam radiation therapy (EBRT)
Severe irritative symptoms	—	Highest risk of worsening (underlying overactivity, poor compliance); short-term relative risk similar across radiation types
Large gland, moderate symptoms	Long-term benefit (avoids progressive LUTS and eventual retention)	Transient worsening → normalisation → improvement as gland regresses

Prostate size

- **American Brachytherapy Society:** $>60 \text{ cm}^3$ technically difficult to implant — **pubic arch interference** and large seed numbers.
- **Cytoreductive androgen-deprivation therapy (ADT)** shrinks the gland by **~25–40%**.
- **$>80 \text{ g}$** → neoadjuvant ADT may be required before radiotherapy, to reduce rectal and bladder dose.
- Unusually large gland with early-stage disease → **surgery may be preferred**, avoiding months of ADT and its effects.

Chronic retention

- **Obstructive retention** → **consider surgery**. Bladder atony → surgery still possible; **urodynamic compliance predicts recovery of continence**.
- Radiation planned → **convert indwelling catheter to intermittent catheterisation** → bladder cycling, preventing radiation fibrosis.

Transurethral resection and radiation

- **Transurethral resection of the prostate (TURP) before, not after, EBRT** for those with or at risk of retention; post-treatment transurethral resection (TUR) → higher incontinence, **especially after brachytherapy**.
- TURP before brachytherapy: historically ↑ post-implant incontinence; contemporary experience challenges this; **limited TUR** (obstructing median lobe/bladder neck, minimal lateral lobe and apical resection) is safe before brachytherapy.
- Previous aggressive TURP → ↑ urinary toxicity with radiotherapy, though long-term severe symptoms may be fewer.
- **Delay radiation 6–8 weeks after TURP** → ↓ incontinence and bladder neck contracture.
- **TURP, microwave or laser defects:** brachytherapy dosimetry challenge (dose is placed peripherally, sparing urethra and bladder neck); **cryoablation** needle placement risks intraurethral position. **Small glands ($<30 \text{ cm}^3$) with large defects** → alternative strategies.

Neurogenic dysfunction

- **Parkinson disease:** careful urological history; urodynamics quantify risk of post-treatment incontinence or worse LUTS.
- **Severe compliance abnormalities** → **surgery preferred**, given progressive bladder fibrosis after radiation.

Concept — *Surgery removes the obstructing gland; radiation leaves it to shrink and scars the bladder* — so obstruction, a large gland and a poorly compliant bladder favour surgery, while irritative symptoms and prior resection defects are worsened or complicated by radiation.

Sexual Function

- **Recovery depends on treatment quality and host factors:** baseline function, use of **phosphodiesterase type 5 (PDE5) inhibitors**, frequency of sexual activity, and vascular risk factors (diabetes, atherosclerosis, hypercholesterolaemia).
- **Age and preoperative erectile function** predict recovery at 2 years after bilateral nerve sparing.
- **Poor baseline function → more non-nerve-sparing surgery**, independent of cancer severity — baseline shapes the plan.
- **Increasing age** → lower potency recovery after surgery and greater decline after radiation.
- **More frequent preoperative sexual activity** → **more postoperative sexual bother**, independent of functional recovery.

Bowel Function/Other Conditions Affecting Treatment Choice

- **Radiotherapy** → **more bowel dysfunction than surgery** in comparative data.
- **Careful gastroenterological history** for radiation candidates: pre-existing faecal frequency/urgency, irritable bowel, tenesmus or extensive rectal surgery → warn of exacerbation; **gastroenterological evaluation before therapy**.
- **Inflammatory bowel disease (Crohn disease, ulcerative colitis)** → suggested more severe acute and late toxicity with pelvic radiotherapy.
- **Neurogenic bladder** → risk of functional deterioration or incontinence after any treatment; neurogenic or inflammatory bowel dysfunction → worse bowel function after EBRT.

CONTRAINDICATIONS AND CAUTIONS BY MODALITY

Modality	Specific contraindication	Relative contraindications / cautions
Radiotherapy	Previous radical pelvic irradiation (history of dose and field needed)	Previous pelvic surgery trapping bowel in the field (radiation enteritis); substantial bowel in the field (e.g. hernia); impaired healing (poorly controlled diabetes); inflammatory bowel disease; pre-existing bowel dysfunction
Surgery	—	Previous pelvic irradiation (↑ complexity of RP); chronic steroid therapy (delayed healing); significant cardiac, pulmonary or haematological comorbidity (anaesthetic, bleeding risk)

Concept — *Radiation's toxicity falls on neighbouring bowel and bladder and depends on tissue healing; anything that places bowel in the field or impairs healing weighs against it.*

Patient Preference/Biases

- **Preference carries unusual weight** because no compelling data favour one treatment in individual cases.
- **Preconceptions** from family, friends, the internet and lay media; many patients consider only surgery.
- **Surgery choosers** → **concerned with cancer mortality**; non-surgical choosers → concerned with quality of life; obstructive LUTS → more likely to elect surgery.
- **Longevity assessment may be the most important component of counselling**, given competing risks and the observed safety of deferral in most favourable- and intermediate-risk cancers.
- Physicians are often **ineffective at conveying patient-specific risks**; patients prefer **specific, quantitative estimates** to generalisations.

- **Shared decision-making:** collaborative decision by patient and clinician, advocated for PSA testing and treatment by the **American Urological Association (AUA)**, **American Society for Radiation Oncology (ASTRO)** and **Society of Urologic Oncology (SUO)**; consensus exists on implementation strategies.
- Patients who **perceive their involvement as active** (whatever their initial preference) → better informed, **less decisional regret**.
- **Decision aids** (written, in-person, web-based): meta-analysis of randomised trials → negligible association with health outcomes; newer trials → improved decision-making. Aids should account for physician recommendation and family wishes.
- **The urologist's recommendation remains the most influential factor.**

Concept — *Without a clearly superior treatment, the decision turns on **which risk domain the patient will accept** — so the quality of information on longevity and side effects, not the length of the menu, determines a good choice.*

ASSESSMENT OF PROSTATE CANCER RISK

Tools for Risk Stratification

- **Purpose:** estimate cancer mortality over time → decide whether to treat and which options are reasonable.
- **Most acknowledged criteria:** clinical stage, biopsy Gleason score, pretreatment PSA.
- **No single criterion** identifies all high-risk patients; **no standardised system** exists — an obstacle to counselling, outcome evaluation, trial design and guidelines.

DEFINITIONS OF LOW, INTERMEDIATE, AND HIGH RISK

- **The core challenge:** apparently indolent cancers with unrecognised aggressive biology.
- **AUA/ASTRO/SUO** stratify into very low, low, intermediate and high; **NCCN additionally substratifies** intermediate and high risk.
- NCCN low- and intermediate-risk groups span a **broad range of aggressiveness**; optimal treatment remains under investigation.
- **Validation of the groups:** very low risk → less adverse surgical pathology than low risk; favourable-risk groups harbour adverse features beyond low risk, affecting surveillance selection; **very high risk → worse pathological and oncological outcomes after RP** and more additional therapy than high risk.

AUA/ASTRO/SUO VERSUS NCCN RISK GROUPS

AUA/ASTRO/SUO	NCCN
Very low	Very low
Low	Low
Intermediate	Favourable intermediate / unfavourable intermediate
High	High / very high

TABLE 158.1 RISK STRATIFICATION AND STAGING EVALUATION

NCCN risk group	Clinical and pathological features	Staging imaging
Very low	All of: T1c; Gleason ≤6/grade group (GG) 1; PSA <10 ng/mL; <3 positive cores with ≤50% cancer in each; PSA density <0.15 ng/mL/g	Not indicated
Low	All of: T1–T2a; Gleason ≤6/GG1; PSA <10 ng/mL	Not indicated

NCCN risk group	Clinical and pathological features	Staging imaging
Favourable intermediate	Intermediate-risk factors (IRF): T2b–T2c, Gleason 3+4=7/GG2, PSA 10–20 ng/mL. Has all of: 1 IRF; GG1 or 2; <50% cores positive (e.g. <6 of 12)	Not indicated
Unfavourable intermediate	IRF as above (GG2 or 3). Has one or more of: 2–3 IRFs; GG3; ≥50% cores positive (e.g. ≥6 of 12)	Technetium-99m methylene diphosphonate (MDP) bone scan + soft-tissue imaging of pelvis, abdomen and chest (chest CT and abdominopelvic computed tomography (CT) or magnetic resonance imaging (MRI))
High	Any of: T3a; Gleason 8/GG4 or 4+5=9/GG5; PSA >20 ng/mL	
Very high	Any of: T3b–T4; primary Gleason pattern 5; >4 cores with Gleason 8–10/GG4–5	

Source: NCCN (Schaeffer et al., 2023).

- **What the table is for:** assigning risk group and deciding when staging imaging begins.
- **Pattern:** favourable groups require **ALL** criteria; adverse groups need **ANY** one. Imaging starts at unfavourable intermediate.
- **Traps:** PSA density appears only in very low risk; GG3 alone makes intermediate risk unfavourable; primary pattern 5 or >4 high-grade cores → very high, not high.

GLEASON SCORING AND GRADE GROUPS

- **Gleason system (1966):** glandular architecture; score = **dominant pattern + largest minority pattern**; patterns 1–5, 5 least differentiated.
- **International Society of Urological Pathology (ISUP) 2005:** patterns 1 and 2 abandoned → effective scale 6–10; several variants formerly pattern 3, and **all cribriform patterns, reclassified as pattern 4**.
- **Grade migration:** more 3+4 and 4+3 than in historical cohorts → better prognostic accuracy, but **historical and contemporary outcomes are not directly comparable**; upward migration of former low-risk cancers has opened surveillance to some Gleason 3+4 disease.
- **Grade groups:** proposed 2013, advocated by ISUP 2014 to communicate aggressiveness on a 1–5 scale; correlation with prostate cancer-specific mortality validated.

GLEASON SCORE TO GRADE GROUP

Grade group	1	2	3	4	5
Gleason score	3+3 (≤6)	3+4	4+3	4+4	4+5, 5+4, 5+5

Concept — The 2005 revision **moved the boundary between patterns 3 and 4**, so today's Gleason 6 is purer than the historical one — improving prognosis while invalidating naive comparison with older series.

EPSTEIN CRITERIA

- **Epstein criteria (1994)** predict **insignificant cancer** — organ-confined (<pT3a), volume <0.5 mL, no high-grade component:
 - **PSA density <0.15 ng/mL/g**
 - **Gleason ≤6**, no adverse biopsy pathology
 - **<3 positive cores** (of a six-core biopsy)
 - **≤50% cancer in any core**
- Predictive accuracy **79%**; revised 2004 with higher accuracy (up to ~84%). **Among the most used tools for selecting surveillance candidates.**

- **Caveat:** validated on **systematic biopsy before surgery**; accuracy with saturation or MRI-targeted biopsy is unclear; modified global grading has been proposed for MRI-targeted biopsy.
- **Other biopsy correlates of aggressiveness and organ confinement:** Gleason score; number of positive cores; distribution and volume of cancer; perineural invasion; lymphovascular invasion; ductal, signet-ring or neuroendocrine differentiation.

STAGING

- **Guidelines vary.** Most support **radionuclide bone scan for high-risk** disease; those addressing soft tissue endorse pelvic CT or MRI.
- **NCCN:** technetium-99m bone scan + CT or MRI for **high-risk and unfavourable intermediate-risk** patients at high risk of occult metastasis (clinical parameters or nomograms). **Equivocal bone scan → positron emission tomography (PET)/CT or MRI.**
- **"Choosing Wisely":** American Society of Clinical Oncology (ASCO) **2012** and AUA **2013** — reduce staging imaging in low-risk disease. Compliance poor: **low-risk over-imaged, high-risk paradoxically under-imaged**; guidelines lower inappropriate imaging but also appropriate imaging.

CONVENTIONAL STAGING MODALITIES

Modality	Role	Strength	Weakness
Bone scan	Standard for initial staging, recurrence and metastatic disease	Easy to interpret, cost-effective, widely available	Low specificity → equivocal results need further imaging; yield depends on PSA, stage, Gleason
CT	Guideline-endorsed for nodal staging ; baseline in grossly advanced local disease with hydronephrosis or retention	Available	Very limited local staging (poor prostate–levator ani separation, poor intraprostatic anatomy); cannot detect change in normal-sized (<10 mm) nodes
MRI	Only examination accurate enough for extraprostatic extension (EPE) and seminal vesicle invasion (SVI)	High spatial and contrast resolution; high positive predictive value (PPV) for EPE	Modest sensitivity and poor negative predictive value (NPV) for EPE; poor for focal EPE

MRI for EPE and SVI

- **T2-weighted imaging (T2WI):** best spatial resolution. **Gross capsular extension or disruption is highly predictive**; subtle irregularity or bulging has low predictive value.
- **Length of capsular contact** improves prediction; adding **diffusion-weighted imaging (DWI)** or **dynamic contrast-enhanced (DCE)** imaging improves subtle EPE detection by localising the dominant tumour — though at lower spatial resolution.
- **High PPV matters in high-risk surgery:** knowing the EPE site helps reduce positive margins.
- **SVI: T2-hypointense mass in the seminal vesicle**, contiguous with tumour at the base; T2WI + DWI improves accuracy; **high sensitivity and very high specificity.**
- EPE and SVI predict **worse oncological outcome and early biochemical recurrence** → counselling, treatment selection and planning.

PSMA PET IN STAGING

- **PET** uses radiolabelled biomolecules to detect metabolic activity or cancer-associated surface molecules; fused with CT or MRI for localisation. Potential roles: **surveillance selection and pre-treatment staging**.
- Focus on **prostate-specific membrane antigen (PSMA) PET** for high-risk staging and detection of recurrence.
- **PSMA PET → greater sensitivity for metastasis than bone scan and CT, while retaining high specificity.**

PSMA LIGAND TRIALS IN HIGH-RISK STAGING

	proPSMA — gallium-68 (Ga-68) PSMA-11 PET-CT	18F-DCFPyL PET/CT (US Food and Drug Administration (FDA)-approved)
Design	Prospective randomised; 302 high-risk; vs conventional imaging	Prospective multicentre phase II/III; 268 high-risk before prostatectomy; vs CT/MRI
Sensitivity	85% vs 38%	40.3% vs 42.6% (similar); 60% for nodes >5 mm
Specificity	98% vs 91%	97.9% vs 65.1%
Other	Accuracy 92% vs 65% ; advantage held for nodal and distant disease; management changed in 27% of conventional arm on crossover PSMA	PPV 86.7% vs 28.3% ; NPV 83.2% vs 77.8%; high affinity for PSMA extracellular domain
Verdict	Appropriate replacement for conventional imaging ; limited US access	Approved on superior PPV ; greatly improved US access

Figure 158.2 — Performance of Ga-68 PSMA PET versus conventional imaging

- Area under the curve, specificity and sensitivity for any metastasis, pelvic nodal and distant disease all favour PSMA PET-CT, and the advantage persists when equivocal lesions are counted as positive (Hofman et al., Lancet 2020 — proPSMA).

GUIDELINES ON MOLECULAR IMAGING IN HIGH-RISK DISEASE

AUA 2022	NCCN 2023
Molecular imaging for high-risk patients with no metastasis on conventional imaging (an add-on)	PSMA PET can be used instead of conventional imaging (a replacement)

Concept — PSMA PET's decisive advantage is **specificity and PPV, not detection of micrometastases**: it confirms what it sees, but a negative scan cannot exclude small nodal disease.

NOMOGRAMS FOR PREDICTION OF STAGE AND DISEASE OUTCOME

- Early PSA era: prognosis rested on **PSA alone**; recognition that Gleason and stage often diverged from PSA → **Partin tables (1993)**: PSA + Gleason + clinical stage → predicted pathological stage at RP.
- **Partin tables**: multi-institutional validation, accurate **72.4%** of the time; generalisable across populations; derived from **Johns Hopkins** RP experience and periodically updated.
- Later organ-confinement nomograms add age, digital rectal examination (DRE), PSA density or percentage core involvement.

RISK STRATIFICATION NOMOGRAMS

- Predict **clinical outcomes after treatment** — relapse and survival.

- Other candidate variables (extent of cancer on biopsy, pre-diagnosis PSA velocity, PSA doubling time, tertiary Gleason pattern) remain of uncertain value — **not routinely collected**.
- Further nomograms inform **active surveillance, RP, neurovascular bundle preservation, omission of pelvic lymph node dissection, brachytherapy and EBRT**; postoperative nomograms (age, PSA, pathological grade and stage) reassess biochemical progression-free survival and the likely success of adjuvant or salvage radiation.

NOMOGRAMS AND RISK SCORES COMPARED

Tool	Inputs	Predicts	Performance
Partin tables	PSA, Gleason, clinical stage	Pathological stage at RP	Accurate 72.4%
Kattan nomogram	PSA, clinical stage, Gleason	Relapse after RP	AUC 0.79
University of California, San Francisco–Cancer of the Prostate Risk Assessment (UCSF-CAPRA)	Age, PSA, clinical stage, biopsy Gleason, % positive cores	Recurrence; later also cancer-specific mortality, independent of treatment	5-year recurrence-free survival 85% (low) → 8% (high)

FIGURE 158.3 UCSF-CAPRA SCORE: COX REGRESSION AND KAPLAN–MEIER RECURRENCE-FREE SURVIVAL (RFS)

CAPRA score	0–1	2	3	4	5	6	≥7
Hazard ratio	Reference	1.28 (NS)	2.36	2.38	3.32	7.11	17.38
5-year RFS	85%	81%	66%	59%	60%	34%	8%

Source: Cooperberg et al., 2005.

- **Pattern:** score 2 is not significantly different from 0–1; 3–5 form a middle plateau; **risk jumps sharply at ≥6**.

Concept — Nomograms rest on **the same three pillars** — **PSA, grade and stage**; they differ chiefly in what they predict: pathological stage (Partin) or post-treatment outcome (Kattan, CAPRA).

PROSTATE MRI IN RISK ASSESSMENT

- **Prostate Imaging Reporting and Data System (PI-RADS) version 2:** minimum technical standards and structured reporting of suspicion for clinically significant cancer.
- **PI-RADS score and apparent diffusion coefficient (ADC) on DWI correlate with outcome** → usable for risk stratification.
- **Surveillance:** tumour ADC may mark progression and identify those who will benefit from radical treatment.
- **Before treatment:** ADC and PI-RADS predict biochemical recurrence after RP and radiation; MRI improves post-EBRT nomograms; pretreatment MRI predicts recurrence after combined brachytherapy + EBRT in intermediate- and high-risk disease.
- **MRI EPE or SVI before RP predicts biochemical recurrence;** substituting MRI for pathological EPE/SVI in UCSF-CAPRA gives **comparable performance** — outcome prediction before surgery.
- With clinical parameters and biomarkers, MRI guides the need for **repeat biopsy when biopsy and MRI are discordant**.

Concept — MRI is moving **from a detection tool to a prognostic variable**: radiological EPE, SVI and ADC can stand in for pathology before any treatment.

GENOMIC TESTS FOR RISK STRATIFICATION

- Clinical risk groups are reliable but **heterogeneous**; genomic tests add **independent prognostic information**, best used alongside stage, Gleason and PSA.

LIQUID VERSUS TISSUE BIOMARKERS

Liquid	Tissue (from biopsy)
Minimal risk — avoid the infection and sepsis risk of transrectal biopsy	Gold standard for representing the tumour environment
Prostate Health Index (PHI); 4Kscore; prostate cancer antigen 3 (PCA3); Mi-Prostate Score; SelectMDx; ExoDx Prostate IntelliScore	ConfirmMDx; Prolaris; Decipher; Oncotype DX Genomic Prostate Score — recommended by the Molecular Diagnostic Services Program (coverage and reimbursement)

GUIDELINES ON GENOMIC TESTING

Body	Position
NCCN	Consider in surveillance candidates with low or favourable intermediate risk and life expectancy >10 years
AUA/ASTRO 2022 (SUO-endorsed)	Not routinely for treatment choice or risk stratification; may be used where extra risk information guides surveillance decisions — high-volume Gleason 3+3 considering treatment, or Gleason 3+4 considering surveillance; not appropriate in small-volume low-risk disease
ASCO	Similar to AUA

- **Limitations:** head-to-head comparisons with clinically relevant end points and cost are needed; results depend on the tissue collected.
- **Intraprostatic genetic heterogeneity** has largely disproven the claim that any sampled focus predicts behaviour; whether **MRI-targeted sampling** of the dominant focus helps is unproven.

Concept — *A genomic score describes only the focus sampled; intraprostatic heterogeneity is why guidelines confine these tests to borderline surveillance decisions.*

SELECTION OF TREATMENT: COMPARATIVE STUDIES

Oncologic Outcomes

- **High-quality comparative data are scarce**; the Institute of Medicine placed localised prostate cancer among the **top 25 comparative-effectiveness priorities**.
- **Biochemical-free survival is a flawed end point:**
 1. Definitions vary widely.
 2. PSA kinetics differ — **radiation's cytotoxic effect lasts months to years and leaves prostatic PSA sources**.
 3. **ADT** (common with high-risk radiation) delays biochemical progression.
- **Cancer-specific and overall mortality are the most valid end points** but need protracted follow-up; enrolling low-risk or comorbid patients → non-oncological deaths obscure treatment effect.

Comparative Population-Based Studies

- Swedish population data, the Cancer of the Prostate Strategic Urologic Research Endeavor (CaPSURE) registry and several Surveillance, Epidemiology, and End Results (SEER) analyses → **higher prostate cancer-specific mortality after radiation than after RP**, adjusted for age and risk.

- **Bias:** historical preference for radiation in older, more comorbid, higher-risk men is incompletely captured; **selection, detection and confounding bias** can yield results contradicting randomised trials.

FIGURE 158.4 RADIOTHERAPY VERSUS RP: CANCER-SPECIFIC MORTALITY (PROPENSITY-ADJUSTED SUBDISTRIBUTION HAZARD RATIO, SHR)

Risk group	sHR (radiotherapy vs RP)	Interpretation
1	1.91 (1.16–3.14)	Favours RP
2	1.77 (1.37–2.29)	Favours RP
3	1.50 (1.19–1.88)	Favours RP
Non-metastatic (1–3)	1.76 (1.49–2.08)	Favours RP
4 (metastatic)	0.76 (0.49–1.19)	No significant difference

Source: Sooriakumaran et al., BMJ 2014. Substratified by age (<65 / ≥65) and Charlson score (0 / ≥1).

- **Pattern:** RP favoured throughout non-metastatic disease, most consistently in **younger and comorbidity-free** subgroups — precisely those most exposed to selection bias.

Scandinavian Prostate Cancer Group Study Number 4 (SPCG-4)

- **RP versus watchful waiting; first randomised intention-to-treat comparison** with overall and cancer-specific survival end points; long follow-up; blinded pathology review.
- **Limitation: pre-PSA era** — nearly 90% of 695 men had palpable disease; does not reflect current practice.
- **23.2 years:** deaths 200/347 (RP) vs 247/348; prostate cancer deaths 63 vs 99 → **relative risk (RR) 0.56;** absolute difference **11.0 percentage points; number needed to treat 8.**
- **Greatest benefit: age <65 (RR 0.45) and intermediate risk (RR 0.38);** older men → fewer metastases (RR 0.68).

Prostate Cancer Intervention Versus Observation Trial (PIVOT)

- **RP versus observation, PSA era;** planned 2000 men, recruited only 731 (addressed by longer follow-up).
- **10 years: no difference in overall or prostate cancer mortality;** no variation by age, race, performance status or comorbidity; **no benefit in low risk;** RP reduced all-cause mortality with **PSA >10 ng/mL** and approached significance in intermediate/high risk; **bone metastases 4.7% vs 10.6%,** more marked with PSA >10 and intermediate/high risk.
- **Limitations:** underpowered (91% power for a 25% relative reduction only after reformulation; ~1200 needed); **~20% non-adherence;** only 10% younger than 60; inclusion required life expectancy >10 years yet **~40% dead at 10 years (~5% from prostate cancer)** and up to 60% at 15 years (~10% from prostate cancer) → poor baseline selection.
- **19.5 years:** absolute risk reduction (ARR) all-cause mortality **5.5% (61.3% vs 66.8%; P = 0.06, NS);** prostate cancer mortality ARR **4.0% (7.4% vs 11.4%; P = 0.06, NS); intermediate risk: all-cause ARR 14.5% (significant);** no benefit in low or high risk. RP reduced treatment for progression but increased incontinence and erectile and sexual dysfunction.

PIVOT SUBGROUP BENEFIT BY DATA CUT

Data cut	Subgroup with all-cause mortality benefit	Reason for difference
10 years	PSA >10 ng/mL (intermediate/high risk approaching significance)	Different follow-up; subgroup defined by PSA versus by risk group
19.5 years	Intermediate risk only	

Prostate Cancer Outcomes Study (PCOS)

- Population-based cohort, age **55–74**, diagnosed 1994–1995; **RP (1164) vs EBRT (491)** within 1 year; propensity-scored multivariable analysis.
- **15 years:** 568 deaths, 104 from prostate cancer; RP → **overall mortality HR 0.60, disease-specific HR 0.35** ($P < 0.0001$).
- Benefit greatest in younger, lower-risk men but also seen in older and high-risk men.
- **Signs of selection bias:** differences appeared **as early as 3 years** — inconsistent with the natural history of early-stage disease; in low risk, overall but not cancer-specific mortality favoured RP → unrecorded differences (volume, true stage, non-oncological mortality predictors).

Prostate Testing for Cancer and Treatment (ProtecT)

- 1999–2009: **82,429 men aged 50–69** PSA-tested; 2664 diagnosed; **1643 randomised** — active monitoring 545, surgery 553, **radiotherapy + 3–6 months ADT** 545. 62% of eligible men accepted randomisation; mostly low–intermediate risk.
- **Adherence:** 88% monitoring, 71% prostatectomy, 74% radiation.
- **10 years (primary end point): no difference in prostate cancer mortality; metastases higher with monitoring.** Mortality ~1% against 15% predicted at design → reduced power.
- **15 years:** prostate cancer mortality **2.7% (surveillance), 2.2% (surgery), 2.9% (radiation);** $P = 0.53$; other-cause death 21.7%; metastatic progression 6.3% overall.
- **Surgery vs radiation:** no difference in mortality, metastasis-free survival or ADT use at 15 years; recurrence and secondary therapy not reported. **Treated at age <65** → lower prostate cancer mortality with surveillance or surgery than radiation; **>65** → no difference surgery vs radiation.
- **Population:** **77% low risk;** only **6% GG ≥3;** 9.6% high risk (D'Amico); 2.5% high risk (UCSF-CAPRA) → explains the low mortality.
- Half of all metastatic or locoregional progression occurred in the surveillance arm; surveillance patients reaching prostatectomy showed **high upgrading and adverse pathology** → poor pre-MRI baseline risk assessment.
- **Verdict:** initial surveillance in favourable-risk disease, including much favourable intermediate risk, **does not greatly affect prostate cancer mortality;** extension to higher risk is unclear.

PROTECT FIGURES THAT DIFFER BY REPORT

Endpoint	10-year report	15-year report	Reason
Surveillance patients treated by 3 years	A quarter	30%	Different data cut
Surveillance patients treated by 10 years	Over half	55%; 60% by 15 years	Different data cut

COMPARATIVE TRIALS AND COHORTS: ONCOLOGICAL OUTCOME

Study	Comparison	Population	Key result	Verdict / caveat
SPCG-4	RP vs watchful waiting (randomised)	Pre-PSA; ~90% palpable	Cancer death RR 0.56; NNT 8 at 23 years	Benefit in <65 and intermediate risk; not today's patients
PIVOT	RP vs observation (randomised)	PSA era; older, comorbid	No mortality difference at 10 years; NS at 19.5 years	Benefit only intermediate risk; underpowered

Study	Comparison	Population	Key result	Verdict / caveat
 ProtecT 	Monitoring vs RP vs RT+ADT (randomised)	Screen-detected, mostly low risk	Cancer mortality ~2–3% in all arms at 15 years	More metastasis on surveillance; <65 favours surgery over RT
 PCOS 	RP vs EBRT (cohort)	Age 55–74, 1994–95	Overall HR 0.60; disease-specific HR 0.35	Early divergence → selection bias
 Swedish, CaPSURE, SEER 	RP vs radiation (population)	Unselected	Higher cancer mortality after radiation	Confounded by allocation bias

Concept — Randomised comparisons with observation are **diluted by competing mortality and low-risk enrolment**, so benefit surfaces only in younger, intermediate-risk men with long follow-up; the consistent observational advantage of surgery over radiation is largely selection bias.

Functional Outcomes

- With oncological differences controversial, **quality of life drives the choice** and must be discussed in detail.
- Older comparisons came from **single-institution centres of excellence** (poor generalisability, selection bias) or population datasets with inadequate fields, instruments and operator variability; large comparative studies now exist (Table 158.2).

TABLE 158.2 PROPENSITY-WEIGHTED PROSTATE CANCER SYMPTOM INDICES SCORES FOR SEXUAL, URINARY, AND BOWEL SYMPTOMS OVER TIME ACROSS DIFFERENT TREATMENT GROUPS AMONG PATIENTS WITH NEWLY DIAGNOSED PROSTATE CANCER

Domain	EBRT	Brachytherapy	Radical prostatectomy
Sexual dysfunction	Worse at 3–12 months, below meaningful threshold; near active surveillance (AS) by 24 months	Worse at 3–12 months, below threshold	Meaningfully worse at 3 and 12 months (+36, +28); still worse at 24 months
Urinary incontinence	No meaningful change	No meaningful change	Meaningfully worse at 3, 12 and 24 months (+34, +18, +15)
Urinary obstruction/irritation	Meaningfully worse at 3 months (+12); resolved by 12 months	Worst acute effect: meaningfully worse at 3 months (+21); resolved by 24 months	Slightly worse at 3 months; better than AS at 12–24 months
Bowel problems	Meaningfully worse at 3 months (+5); still worse at 24 months	No meaningful change	No change

Scores 0–100, higher = worse; figures are mean difference versus AS. Clinically meaningful = exceeding 0.5 standard deviation of the AS baseline score. Among AS patients, 18.7% were treated within 24 months. Source: Chen et al., JAMA 2017.

- **What the table is for:** each modality's **signature domain and time course**, measured against surveillance.
- **Pattern:** RP → incontinence and sexual dysfunction, early and persistent; EBRT → acute bowel and irritative symptoms; brachytherapy → the most severe acute obstructive/irritative symptoms.
- **Traps:** brachytherapy, not EBRT, has the worst early urinary irritation; RP **improves** obstructive/irritative scores by 1–2 years.

PCOS

- RP → **more incontinence and erectile dysfunction at 2 and 5 years; less bowel urgency at 2 and 5 years; no significant difference in any domain at 15 years.**
- Convergence reflects decline in both groups: age, declining sexual interest or acceptance of dysfunction, and additional cancer treatment.

 ProtecT

- **Prostatectomy: greatest harm to sexual function and continence;** partial recovery but worse than other arms throughout.
- **Radiotherapy:** sexual function worst at 6 months, then partial recovery, stable below baseline; **little effect on continence;** bowel function worse at 6 months then recovered, **except increasing bloody stools;** voiding and nocturia worse at 6 months, equal by 12 months.
- **Active monitoring:** gradual decline in sexual and urinary function.
- **No differences** in anxiety, depression, or general or cancer-related quality of life.

Comparative Effectiveness Analysis of Surgery and Radiation (CEASAR)

- Prospective population-based study, enrolled 2011–2012.
- **Low risk: 3-year sexual function score 12 points better after EBRT than RP (52 vs 39);** high risk: difference not clinically significant; with a **binary definition** (erections firm enough for intercourse), no major difference in any risk stratum.
- With AS added: RP → greater decline in sexual function and continence than EBRT or AS, but **fewer irritative symptoms than AS;** no meaningful bowel or hormonal differences beyond 12 months.
- **Erectile disadvantage of RP seen only in the highest quartile of baseline function;** none in the remaining three quartiles.

FUNCTIONAL OUTCOME STUDIES COMPARED

Study	RP	Radiation	Long-term
PCOS	More incontinence, erectile dysfunction at 2–5 years	More bowel urgency at 2–5 years	No difference at 15 years
ProtecT	Worst sexual function and continence throughout	Transient bowel, voiding and sexual effects; rising bloody stools	No difference in anxiety, depression or quality of life
CEASAR	Worse sexual function (low risk) and continence at 3 years	Better 3-year sexual scores in low risk	Difference confined to best baseline function

Open Versus Robotic Prostatectomy

- **One randomised phase III trial:** robotic-assisted laparoscopic and open RP → **similar functional outcomes at 12 weeks.**
- Robotic surgery improves functional and oncological outcomes **along a single surgeon's learning curve.**
- Community data: similar recovery pattern over 3 years → **patients should not expect different oncological or quality-of-life outcomes by approach.**
- Robotic surgery may ease identification of nerve-preservation planes and wider dissection in pT3 disease — possibly **surgeon bias.**

Concept — Surgery front-loads urinary and sexual harm that partly recovers; radiation spares early function but adds bowel toxicity and a slow cumulative decline — the curves converge by 15 years, so the choice is about when and where the patient accepts the harm.

PATIENT COUNSELLING: A STEPWISE DECISION APPROACH

- Many options and conflicting advice → **confusion and anxiety**; a menu of options lacks context because the goals of the strategies diverge.
- **A methodical, stepwise approach** reduces confusion; the patient decides **in which domain to accept risk** — oncological control and cancer mortality versus side effects and quality of life.

FIGURE 158.5 PATIENT COUNSELLING IN LOCALISED PROSTATE CANCER: A STEPWISE DECISION APPROACH

Decision	Inputs	Branches
1. Treatment vs no treatment	Disease: grade, stage, genomics, family history. Host: age, comorbidity, anticipated longevity, preference	No treatment → observation (watchful waiting or AS)
2. Partial vs whole gland	Disease: grade, stage, focality, tumour size. Host: age, potency, urinary symptoms, preference	Partial → ablative therapy (mapping method, tumour location, energy source)
3. Surgery vs radiation vs whole-gland ablation	Shared decision-making: expected outcomes and toxicities; impact on baseline quality of life; risk of recurrence/secondary treatment	→ Decision 4
4. Specific treatment	Surgery: robotic vs open; lymph node dissection; choice of surgeon. Radiation: external vs interstitial; fractionation; use of ADT	Also applies to ablative therapy

Decision One: Observation Versus Active Treatment

- **The most important decision**; requires risk stratification, education and informed shared decision-making.

ACTIVE SURVEILLANCE VERSUS OBSERVATION

	Active surveillance	Observation (watchful waiting)
Method	Active monitoring by repeated testing	Passive monitoring for symptomatic progression
Goal	Defer treatment until clear need; curative intent retained	Avoid treatment ; accept risk of progression and death
Candidate	Good longevity, few comorbidities, low clinical risk	Low risk of cancer death because of competing risks and long lead time
Follow-up	Scheduled PSA, biopsy, imaging; adherence essential	Periodic PSA, examination, selective imaging
Treatment trigger	Predetermined signs of progression → curative therapy	Symptoms or extensive metastasis → ADT mainstay ; palliative radiotherapy/surgery for disease refractory to systemic therapy

- In practice many are monitored by a **mixture without a clear goal**; there is no consensus on optimal monitoring.
- Surveillance safety is well shown long-term, but patients must **accept uncertainty in exchange for quality of life**, adhere to follow-up and accept treatment at progression; **anxiety → consider treatment**.
- **Intermediate- or high-risk disease → strongly discourage surveillance**, though age and comorbidity may permit it.
- **AUA, ASTRO, SUO and NCCN endorse surveillance in low risk**, and its use is rising: **best option for very low risk; preferable for most low risk**.

- **Limitations:** 25–45% occult high-grade cancer in low-grade disease on systematic biopsy; no consensus on monitoring; declining adherence; unclear intervention triggers.
- Outcomes remain favourable because most occult upgrading is to **GG2 (3+4)**, of low metastatic potential in the first decade → interest in **low-volume GG2 surveillance**, tempered because **any pattern 4 raises eventual metastatic risk**.
- **ProtecT at 15 years** (mostly low or favourable intermediate risk): no mortality difference despite more metastasis → deferral relatively safe in favourable risk.
- **Goals of surveillance:** candidate selection → monitoring for progression → judicious treatment.
- **Refining selection:** MRI and MRI-targeted biopsy now in many algorithms; most guidelines use MRI selectively, but **the National Institute for Health and Care Excellence (NICE) mandates MRI for all surveillance patients**. Genomic testing informs occult adverse pathology and progression; high MRI or genomic scores → upgrading on biopsy.
- **Predictors of reclassification:** core cancer density and PSA density >0.15; Black race and age do not predict it.

INCLUSIVE (TORONTO) VERSUS RESTRICTIVE (JOHNS HOPKINS) SURVEILLANCE

	Toronto — inclusive	Johns Hopkins — restrictive
Entry	All low risk + selected intermediate risk (Gleason 7 or PSA >10)	Epstein criteria: Gleason 6, ≤2 positive cores, no core >50%, PSA density <0.15
15-year cancer mortality	5.7% (actuarial); mature cohort 1.5%	0.6%
Treatment rate	25%, 36%, 45% at 5, 10, 15 years	50%, 57% at 10, 15 years (grade progression 26%, 31%)
Other	Median follow-up 9 years: non-prostate death 10× prostate cancer death; 15% dead by 15 years; 2.8% metastasis; most metastatic cases were Gleason 7 at diagnosis	Stricter entry yet higher treatment rate

- **Emerging consensus: the right strategy lies between the two extremes.**
- **Death on surveillance is mostly cardiovascular**, rarely prostate cancer.
- Benefit of treating **PSA progression alone** is unclear; outcomes support safety, but high treatment rates demand adherence.

TABLE 158.3 SELECTION CRITERIA FOR ACTIVE SURVEILLANCE

Programme	Clinical stage	PSA	Biopsy Gleason	PSA density	Positive cores
University of Toronto	T1c/T2a	≤10–15 ng/mL	≤3+3=6	—	—
Prostate Cancer Research International Active Surveillance (PRIAS)	T1c/T2a	≤10 ng/mL	≤3+3=6	≤0.2 ng/mL/cm ³	≤2
Johns Hopkins	T1c only	—	≤3+3=6	≤0.15 ng/mL/cm ³	≤2
Canary Collaboration	T1/T2	Any	≤7	Any	Any

Sources: Klotz et al., 2015 (Toronto); Bokhorst et al., 2015 (PRIAS); Tosoian et al., 2015 (Johns Hopkins); Newcomb et al., 2016 (Canary).

- **Pattern:** strictness Hopkins > PRIAS > Toronto > Canary; **only Canary admits Gleason 7**; only PRIAS and Hopkins use PSA density.
- **Trap:** Hopkins is the only programme restricted to **T1c**.

Method of surveillance

- **Serial annual biopsy** (traditional) detects occult high grade and progression but **cannot distinguish them**.
- **Confirmatory biopsy within 6–12 months of diagnosis** in most protocols; later frequency varies; modelling shows **less frequent biopsy only slightly delays detection** while greatly reducing biopsies.
- **PRIAS** (increasingly used, especially in Europe): periodic PSA with less frequent biopsy; **PSA doubling time (PSADT)** triggers biopsy or intervention.
- **MRI-led monitoring**, biopsy only on clinical or radiological progression → still **investigational**.

TABLE 158.4 NATIONAL ACTIVE SURVEILLANCE GUIDELINES

Body	Low risk	Intermediate risk	Monitoring	Other tests
ASCO	AS preferred	Active treatment; AS for select patients	PSA every 3–6 months; DRE annually; systematic biopsy within 6–12 months, then every 3–5 years	Other tests investigational
NCCN	Very low: AS preferred. Low: AS preferred for most; all therapies are options	AS and all therapies are options	PSA no more often than every 6 months unless indicated; biopsy no more often than annually	Confirmatory testing within 6–12 months strongly recommended
AUA	Very low and low: AS preferred	Active treatment; AS for select patients	Serial PSA and repeat biopsy	MRI augments, does not replace, surveillance biopsy
NICE	AS preferred	AS or active treatment	MRI at 12–18 months	Concern over clinical or PSA change → MRI and/or rebiopsy

Sources: Bekelman et al., 2018 (ASCO); Schaeffer et al., 2023 (NCCN); Eastham et al., 2022 (AUA); NICE NG131.

- **Pattern**: all agree on **surveillance for low risk**; they diverge on **intermediate risk** (NCCN most permissive) and on **MRI** (NICE MRI-led; AUA MRI as adjunct to biopsy).

Concept — Surveillance is **deferral, not avoidance**: its safety depends on accurate entry selection and adherence, which is why entry criteria, MRI and confirmatory biopsy dominate the debate.

Decision Two: Partial-Gland Versus Whole-Gland Treatment

- Offered where focal therapy is available — typically **thermal ablation**. Selection is controversial but usually requires **unifocal or limited-focality cancer with spatial concordance between imaging and biopsy**.
- **Counselling**: investigational; no long-term follow-up; possible retreatment; **worse outcomes if conventional salvage therapy is later needed**. Ideally within a **clinical trial**.
- **Selection** by transperineal template mapping biopsy or by imaging. **High risk: consensus against** (no long-term locoregional control data). Low risk (usually suited to surveillance): no consensus. **Most favour intermediate-risk, limited-focality disease**. A joint FDA, SUO and AUA workshop addressed trial design, end points and selection.
- **Unilateral disease on systematic biopsy is usually not unilateral at prostatectomy**; low-risk clinical features do not improve prediction. Template mapping better shows multifocality (optimal template undecided); **MRI-targeted biopsy** improves selection, accepting untreated indolent foci.
- **Index lesion hypothesis**: the dominant tumour — generally the **largest focus with the highest Gleason score, most likely MRI-visible** — drives metastasis and lethality. Untested long term, yet used to justify

leaving low-grade secondary and tertiary foci untreated → **high likelihood of residual cancer**; long-term efficacy and survival impact unknown.

Concept — Focal therapy rests on the **untested index-lesion hypothesis**, which is why it is confined to limited-focality intermediate-risk disease and, ideally, to trials.

Decision Three: Surgery Versus Radiation Versus Whole-Gland Ablation

- **Whole-gland ablation** shares focal therapy's problems: little long-term data, frequent retreatment, and **salvage surgery complicated by periprostatic fibrosis**. Its attraction is **preservation of sexual and urinary function**.

WHOLE-GLAND CRYOABLATION VERSUS HIGH-INTENSITY FOCUSED ULTRASOUND (HIFU)

Cryoablation	HIFU
Erectile dysfunction almost always	Function preservation more common , varying with aggressiveness and oncological efficacy; retreatment 29–38% , with more erectile dysfunction and stricture after a second treatment

- **Most whole-gland decisions are surgery versus radiation**: assess longevity, baseline function and goals. Radical-therapy patients prioritise **durable cancer control**; risk models inform efficacy and the likelihood of secondary or multimodal treatment; **longevity is most important for discussing durability**.
- **ProtecT 15 years**: no difference in mortality or metastasis between surgery and radiation; recurrence differences less clear; **treatment before 65 → lower prostate cancer mortality with surgery than radiation**. Unless the patient lives a very long time, surgery's perceived durability may not change longevity.
- Radiation forms share broadly **comparable side-effect profiles**; oncological outcomes are **relatively equivalent at 10 years**; the choice mostly rests on **which anticipated side effects the patient finds most tolerable**.

COUNSELLING FOR SURGERY VERSUS RADIATION

Surgery	Radiation
Early incontinence and impotence are the norm → prepare the patient to avoid decisional regret; give a realistic recovery timeline and plan	Avoids short-term urinary and sexual effects
Quote surgeon-specific outcomes — results are highly surgeon-dependent	Incremental yearly decline in urinary and sexual function beyond the matched population
Significant baseline obstructive/irritative symptoms → may benefit despite early urinary deterioration	Secondary rectal and bladder malignancy at 15–20 years (controversial)
Limited 15–20-year data → counsel younger men within their estimated longevity	Intermediate/high risk → concomitant ADT , which drives much early toxicity; caution young patients about long-term recurrence and toxicity

Concept — Surgery versus radiation is **a trade in the timing of harm** — early, partly recoverable functional loss against avoided early harm but cumulative late toxicity — which is why the young patient's longevity tips the balance.

Decision Four: Type of Treatment Delivery

- Best discussed by a **specialist experienced in that modality**.
- **Surgery**: open versus robotic-assisted laparoscopic approach, chosen by the surgeon's experience and comfort.

- **Radiation options:** EBRT (intensity-modulated, image-guided and/or three-dimensional conformal); interstitial (**permanent seed implants or high-dose-rate temporary insertion**); stereotactic body radiotherapy; proton therapy; combinations; conventional or **hypofractionated** schedules.
- Concern that counselling is biased by **financial incentives and physician bias**.

CONCLUSION

- Active management depends on **evaluation of both the cancer and the host**: relative risk of progression, metastasis and cancer death framed against **competing mortality and expected longevity**.
- **Cancer risk:** biopsy pathology, image-based staging, MRI, genomic markers — combined in validated nomograms or risk scores. **Longevity:** age, comorbidity and standardised scales.
- Because treatments are many and their goals divergent, a **stepwise decision-tree approach** simplifies counselling, informs decisions and reduces decisional regret.

KEY POINTS

- **Individualised assessment** of prostate cancer mortality risk against competing risks of death over time.
- Modality selection by **shared decision-making**: risks and benefits, host factors and patient preference.
- **Baseline urinary, bowel and sexual function** determine the quality-of-life benefit and risk of each treatment.
- **Longevity assessment** may be the most important part of counselling, given competing risks and the safety of deferral in most favourable- and intermediate-risk cancers.
- Mortality risk is assessed with **biopsy pathology, genomic testing, imaging and nomograms**.
- A **methodical stepwise approach** to the patient and the options reduces confusion and anxiety.